

Does clinical context help a chest-radiograph pneumonia model?

A time-safe multimodal study with multi-seed reproducibility and missing-data cautions

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Abstract

Emergency-department (ED) triage produces structured clinical data before chest imaging, raising the question of whether fusing that context with the chest radiograph (CXR) improves automated pneumonia detection. We build a temporally safe multimodal cohort from MIMIC-CXR-JPG, MIMIC-IV and MIMIC-IV-ED, anchoring every feature at the CXR acquisition time t_0 and admitting only measurements resulted at or before t_0 . On a patient-level temporal split (test $n = 1,075$, prevalence 45.3%), a fine-tuned DenseNet-121 reaches AUROC 0.737 ± 0.003 across five training seeds, far above triage-only baselines (logistic regression 0.606, XGBoost 0.611). Adding triage vitals leaves discrimination unchanged ($\Delta\text{AUROC} +0.004 \pm 0.009$, within a pre-specified ± 0.05 non-inferiority margin). Two reproducibility findings dominate. First, a calibration advantage reported from a single checkpoint (expected calibration error, ECE, 0.040 vs 0.067) shrinks to $\Delta\text{ECE} -0.013 \pm 0.016$ across seeds—single-seed evaluation overstated the effect roughly twofold. Second, adding a broad early-lab panel produces a small, seed-consistent AUROC gain ($+0.009 \pm 0.005$, 5/5 seeds), but an ablation shows this gain is reproduced exactly by the lab *missingness indicators alone*, with no contribution from the lab values—the apparent benefit is missing-data structure, not chemistry, and is not attributable to leakage (temporal, split and preprocessing audits all clean). On external validation against 112,120 NIH ChestX-ray14 radiographs, discrimination transfers with minimal loss (AUROC 0.722 ± 0.005 , -0.016 vs internal). We further report subgroup performance gaps not closed by fusion, that a modern CXR foundation backbone does not outperform the task-tuned model, and that CRP and procalcitonin are available before imaging in under 1% of studies. The chest radiograph carries the discriminative signal for radiographic pneumonia; immediately available clinical context does not reliably add discrimination, and naive multimodal fusion can appear beneficial for reasons that will not generalise.

Introduction

Pneumonia is a leading cause of ED presentation and admission, and the chest radiograph remains central to its diagnosis [1]. Deep-learning CXR classifiers approach radiologist-level performance on several findings, and a substantial literature proposes fusing imaging with structured clinical data [2–4]. Three methodological issues recur in that literature and motivate this study. First,

temporal safety: clinical variables are frequently included without verifying they were available at prediction time, risking optimistic leakage. Second, **single-seed reporting:** calibration and small discrimination effects are often reported from one training run, although both are sensitive to training stochasticity [5, 6]. Third, **missing data:** clinical inputs in the ED are heavily and non-randomly missing, so a model can exploit *whether* a measurement exists rather than its value. We address a deliberately conditional question—*when, if ever, does immediately available clinical context help a CXR pneumonia model?*—under strict time-safety, with multi-seed replication and explicit missing-data and leakage audits. Our contribution is not a new architecture but a rigorous, reproducible evaluation and two transferable cautions for clinical multimodal fusion, reported in line with prediction-model and medical-imaging guidance [7, 8].

Results

The image dominates; fusion is discrimination-neutral

Across five seeds (test $n = 1,075$; Table 1, Fig. 1), both CXR models beat triage-only baselines by more than 12 AUROC points. Adding triage vitals does not change discrimination (concat – image $\Delta\text{AUROC} +0.004 \pm 0.009$; lower bound of the effect well inside the -0.05 non-inferiority margin [9]). Attention fusion matches image AUROC but is severely and variably miscalibrated (ECE 0.076 ± 0.042).

Table 1: Multi-seed test-set performance (mean $\pm$ SD over 5 seeds; $n = 1,075$; prevalence 45.3%). Clinical baselines are single-fit.

Model	AUROC	AUPRC	ECE	Brier
Image-only DenseNet-121	0.737 ± 0.003	0.719 ± 0.004	0.053 ± 0.008	0.207 ± 0.001
Multimodal — concat (triage)	0.741 ± 0.006	0.715 ± 0.005	0.040 ± 0.014	0.205 ± 0.002
Multimodal — attention (triage)	0.738 ± 0.006	0.713 ± 0.012	0.076 ± 0.042	0.212 ± 0.010
Multimodal — concat (+labs)	0.747 ± 0.004	0.724 ± 0.004	0.043 ± 0.011	0.203 ± 0.002
Logistic regression (triage)	0.606	0.548	0.037	0.242
XGBoost (triage)	0.611	0.567	0.046	0.241



Figure 1: Per-model AUROC and ECE across five seeds (points, seeds; red bar, mean). Discrimination is stable; the image-only ECE alone varies substantially by seed.

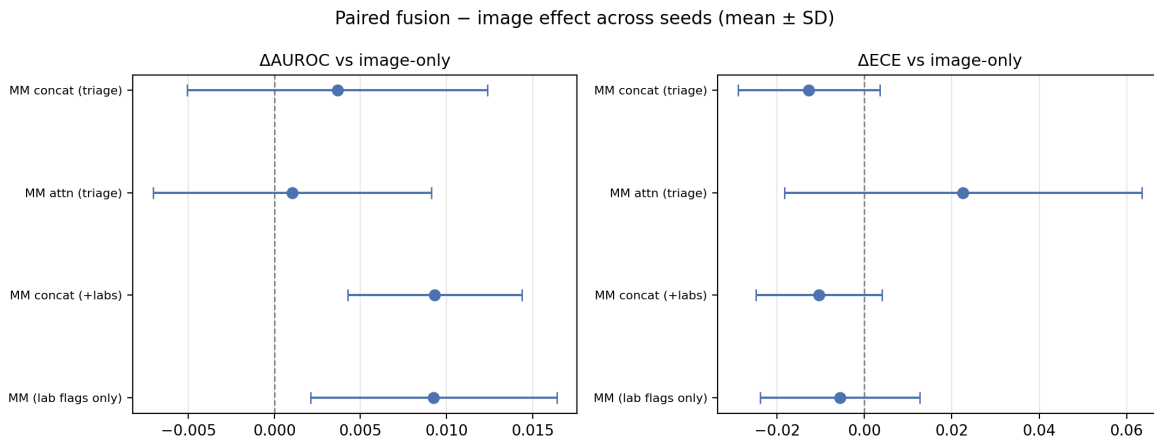


Figure 2: Paired fusion – image effects across seeds (mean $\pm$ SD). Every effect straddles zero; the labs and lab-flags-only deltas are identical, indicating the labs effect is driven by missingness rather than lab values.

Reproducibility caution 1: single-seed evaluation inflates the calibration effect

A single checkpoint suggested the concat model was markedly better calibrated (ECE 0.040 vs 0.067, a 40% reduction). Across seeds the image-only ECE alone ranges 0.041–0.060, and the paired advantage shrinks to $\Delta\text{ECE } -0.013 \pm 0.016$ (4/5 seeds favour concat; range crosses zero). The single-seed estimate overstated the benefit roughly twofold. The effect is also binning-dependent (Fig. 6; $\Delta\text{ECE } -0.005$ at 5 bins to -0.013 at 10 bins [6]). Brier scores are essentially unchanged, so any effect is distributional alignment, not refinement.

Reproducibility caution 2: the labs “benefit” is missing-data structure

Appending early labs gives the only seed-consistent AUROC gain (labs – image $+0.009 \pm 0.005$, 5/5 seeds, range excludes zero). Three analyses show this is not a genuine chemistry signal and not leakage. *Ablation:* a model using lab *missingness flags only* (no values) reproduces the gain exactly (flags – image $+0.009$; Fig. 2); the lab values add nothing further to discrimination. *Subcohort:* on the 269 test studies that actually have labs, discrimination is tied (labs 0.778 vs image 0.776). *Leakage audit:* 0 of 314,145 lab measurements post- t_0 ; 0 cross-split subjects; preprocessing fit on train only; the “any lab drawn” indicator has near-null univariate AUROC (0.464). The gain therefore reflects the weak complementary signal of *whether* bloodwork was ordered, is small, and lies within the patient-level bootstrap CI—a concrete caution that multimodal fusion with heavily missing inputs can appear beneficial for reasons that will not transfer.

Clinical availability of labs

Under time-safe matching, only 25.4% of ED-anchored CXR studies (272/1,075 test) have any resolved lab at or before imaging; CBC and metabolic panels reach $\sim 25\%$, but CRP and procalcitonin—the most pneumonia-specific markers—are present in 0.5% and 0.6% (Fig. 5). At the moment of imaging, the informative labs are essentially unavailable.

Operating points, subgroups, and modern backbone

At a fixed sensitivity of 0.90, specificity is 0.31 (image) vs 0.32 (concat); at 0.95, 0.17 vs 0.18—clinically equivalent. Subgroup AUROC reveals disparities not closed by fusion (Fig. 4): male 0.77

vs female 0.71; White 0.75, Black 0.71, Hispanic 0.64 (small strata, wide CIs); PA view 0.75 vs AP 0.72 [10, 11]. A CheXpert-trained DenseNet-121 [12], used zero-shot (AUROC 0.667) or as a frozen-feature linear probe (0.687), does not outperform our task-tuned model (0.737).

External validation

We evaluated the image models on all 112,120 radiographs of NIH ChestX-ray14 [13] (a different institution, scanner population and label pipeline; 1,431 pneumonia-positive, prevalence 1.28%). Discrimination transfers with minimal loss: AUROC 0.722 ± 0.005 across five seeds (ensemble 0.726), only 0.016 below the internal 0.737 (Fig. 3). As expected under the large prevalence shift (45% $\rightarrow$ 1.3%), prevalence-dependent metrics degrade (AUPRC 0.030, ECE 0.44) because probabilities are calibrated to the internal base rate; AUROC is the appropriate prevalence-independent cross-dataset metric, and it holds despite domain shift and NLP-derived external labels.

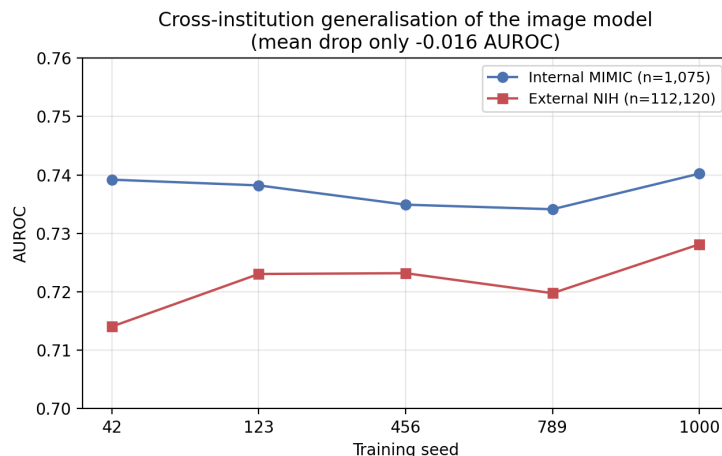


Figure 3: Cross-institution generalisation: per-seed AUROC on internal MIMIC ($n = 1,075$) vs external NIH ChestX-ray14 ($n = 112,120$); mean drop -0.016 .

Discussion

The chest radiograph carries the discriminative signal for radiographic pneumonia in this ED cohort; immediately available structured clinical context—triage vitals, and even a broad early-lab panel—does not add reliable discrimination, and the image model generalises externally with minimal loss. This is a well-powered negative answer to a specific, clinically motivated question, accompanied by two transferable methodological cautions. Reporting calibration or small discrimination effects from a single training run can overstate them (here $\sim 2\times$); multi-seed replication should be standard. And where clinical inputs are heavily and non-randomly missing, a multimodal model can exploit missingness structure and present a spurious benefit that neither reflects the measured values nor generalises—a failure mode distinct from data leakage, which our audits explicitly excluded.

Limitations. Training and multimodal evaluation are single-institution (MIMIC) and retrospective; external validation confirms the image branch generalises to NIH but is necessarily limited to imaging, as no external dataset links ED triage to the CXR, and probabilities require recalibration

to the target base rate before deployment (the `u_ignore` policy also raises internal evaluation prevalence to 45.3%). The lab-present subcohort is small ($n = 269$); the modern-backbone comparison uses frozen features rather than end-to-end fine-tuning; and NIH labels are NLP-derived, adding label noise to the external estimate.

Methods

Data and cohort

We link MIMIC-CXR-JPG v2.1.0 [14], MIMIC-IV v2.2 [15] and MIMIC-IV-ED v2.2 [16]. The anchor t_0 is the CXR acquisition timestamp (DICOM StudyDate + StudyTime). Studies are restricted to frontal ED-anchored radiographs with a linkable ED stay. Pneumonia labels are the CheXpert label [17] derived from the radiology report; radiologically uncertain studies are excluded (`u_ignore`). The cohort comprises 9,154 ED-anchored studies; the patient-level temporal split assigns train 7,144 / validation 930 / test 1,080 studies, and models train and evaluate on radiographs available on disk, yielding a held-out test set of $n = 1,075$ (487 positive / 588 negative; prevalence 45.3%).

Temporal safety and split

Every clinical feature is a triage measurement or a laboratory value resolved at or before t_0 ; for labs we additionally require charttime within $(t_0 - 24\text{ h}, t_0]$. We verified time-safety empirically: of 314,145 extracted lab measurements, zero have charttime after t_0 . Patients are ordered by their first t_0 and assigned 80/10/10 at the patient level; no patient appears in more than one split (0 cross-split subjects, 0 duplicate studies).

Models

The image model is a DenseNet-121 [18], ImageNet-initialised, multilabel-pretrained on 182,637 non-ED MIMIC-CXR studies, then fine-tuned for binary pneumonia (224 px, class-balanced BCE). Multimodal variants concatenate image features with a tabular MLP over triage features (vitals, acuity, view, demographics, per-vital missingness flags), or fuse them by a small Transformer (attention ablation). The +labs variant appends 26 early-lab values and their missingness flags. Clinical baselines are logistic regression and XGBoost [19] on triage features. A modern-backbone baseline uses a CheXpert-trained DenseNet-121 [12] zero-shot and as a frozen-feature linear probe. All deep models share the pretrained backbone so multi-seed analysis isolates fine-tuning stochasticity; tabular preprocessing (median imputation, standardisation, one-hot) is fit on train only.

Evaluation

Primary metrics are AUROC, AUPRC, ECE (10 uniform bins) [5] and Brier score. Each deep model is trained across five seeds {42, 123, 456, 789, 1000}; we report mean $\pm$ SD and paired per-seed deltas against the image-only model on identical test rows. Within-checkpoint uncertainty uses a 2,000-replicate patient-level paired bootstrap. A non-inferiority margin $\Delta_0 = -0.05$ AUROC was pre-specified. Additional analyses: ECE bin-count/scheme sensitivity, operating points at fixed sensitivity, subgroup metrics, a missingness-only labs ablation, and external validation on all NIH ChestX-ray14 radiographs [13] with preprocessing matched to the internal eval transform.

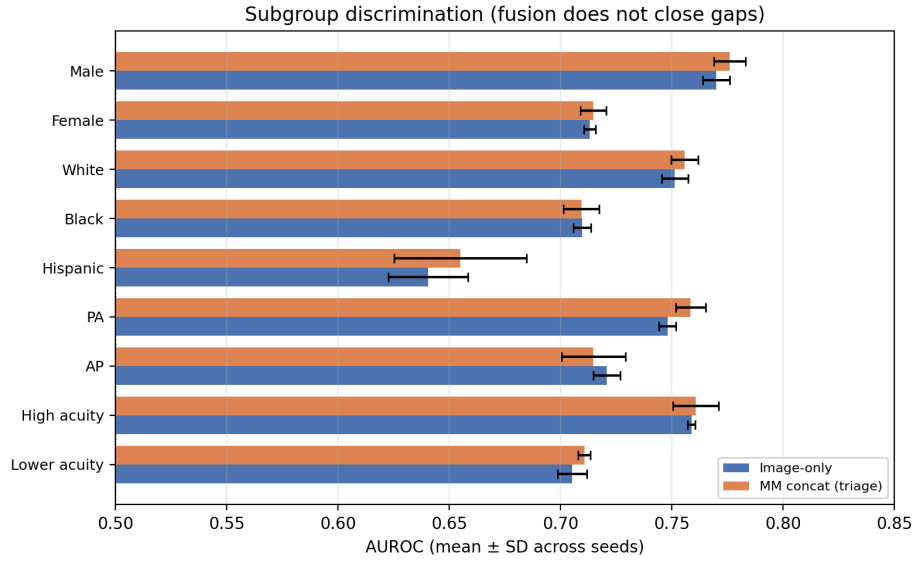


Figure 4: Subgroup AUROC (mean $\pm$ SD across seeds) for image-only vs concat fusion; fusion does not close sex, race or view gaps.

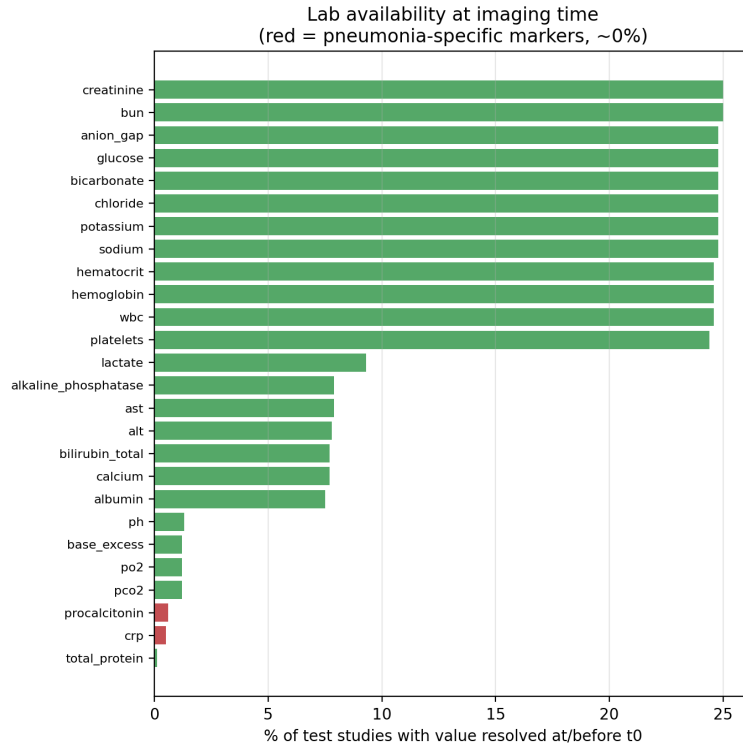


Figure 5: Fraction of test studies with each lab resolved at or before t_0 ; pneumonia-specific markers (red) are essentially unavailable at imaging time.

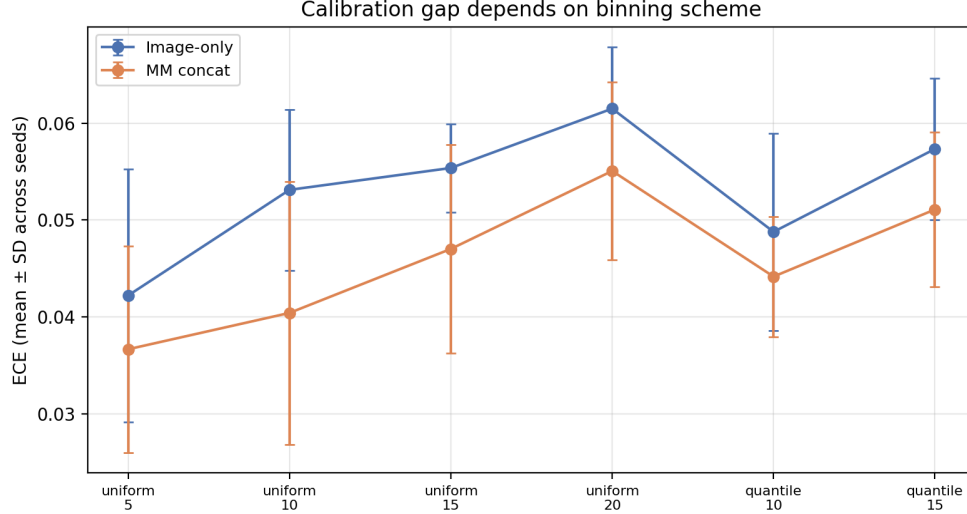


Figure 6: ECE under different binning schemes; the image-concat calibration gap depends on the binning choice.

Data and code availability

All results derive from scripts in the project repository, tied to the named temporal split: cohort construction (`scripts/run_data_pipeline.sh`), multi-seed training and aggregation (`scripts/multiseed/`) and analyses (`scripts/analysis/`). MIMIC and NIH ChestX-ray14 data are publicly available (MIMIC via PhysioNet credentialing); no patient-level data are redistributed.

References

- [1] Joshua P Metlay, Grant W Waterer, Ann C Long, et al. Diagnosis and treatment of adults with community-acquired pneumonia (ats/idsa guideline). *American Journal of Respiratory and Critical Care Medicine*, 200(7):e45–e67, 2019. doi: 10.1164/rccm.201908-1581ST.
- [2] Julián N Acosta, Guido J Falcone, Pranav Rajpurkar, and Eric J Topol. Multimodal biomedical AI. *Nature Medicine*, 28(9):1773–1784, 2022. doi: 10.1038/s41591-022-01981-2.
- [3] Nasir Hayat, Krzysztof J Geras, and Farah E Shamout. MedFuse: Multi-modal fusion with clinical time-series data and chest X-ray images. In *Machine Learning for Healthcare (MLHC)*, volume 182 of *PMLR*, pages 479–503, 2022. URL <https://proceedings.mlr.press/v182/hayat22a.html>.
- [4] Luis R Soenksen, Yu Ma, Cynthia Zeng, Leonard Boussieux, Kimberly Villalobos Carballo, Liangyuan Na, Holly M Wiberg, Michael Lingzhi Li, Ignacio Fuentes, and Dimitris Bertsimas. Integrated multimodal artificial intelligence framework for healthcare applications. *npj Digital Medicine*, 5:149, 2022. doi: 10.1038/s41746-022-00689-4.
- [5] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *ICML*, volume 70, pages 1321–1330, 2017. URL <https://proceedings.mlr.press/v70/guo17a.html>.

- [6] Jeremy Nixon, Michael W Dusenberry, Linchuan Zhang, Ghassen Jerfel, and Dustin Tran. Measuring calibration in deep learning. In *CVPR Workshops*, 2019. URL <https://arxiv.org/abs/1904.01685>.
- [7] Gary S Collins, Karel G M Moons, Paula Dhiman, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, 2024. doi: 10.1136/bmj-2023-078378.
- [8] Ali S Tejani, Michail E Klontzas, Anthony A Gatti, John T Mongan, Linda Moy, Seong Ho Park, and Charles E Kahn. Checklist for Artificial Intelligence in Medical Imaging (CLAIM): 2024 update. *Radiology: Artificial Intelligence*, 6(4):e240300, 2024. doi: 10.1148/ryai.240300.
- [9] Gilda Piaggio, Diana R Elbourne, Stuart J Pocock, Stephen J W Evans, and Douglas G Altman. Reporting of noninferiority and equivalence randomized trials: Extension of the CONSORT 2010 Statement. *JAMA*, 308(24):2594–2604, 2012. doi: 10.1001/jama.2012.87802.
- [10] Laleh Seyyed-Kalantari, Haoran Zhang, Matthew B A McDermott, Irene Y Chen, and Marzyeh Ghassemi. Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations. *Nature Medicine*, 27:2176–2182, 2021. doi: 10.1038/s41591-021-01595-0.
- [11] Agostina J Larrazabal, Nicolás Nieto, Victoria Peterson, Diego H Milone, and Enzo Ferrante. Gender imbalance in medical imaging datasets produces biased classifiers for computer-aided diagnosis. *PNAS*, 117(23):12592–12594, 2020. doi: 10.1073/pnas.1919012117.
- [12] Joseph Paul Cohen, Joseph D Viviano, Paul Bertin, Paul Morrison, Parsa Torabian, Matteo Guarrera, Matthew P Lungren, Akshay Chaudhari, Rupert Brooks, Mohammad Hashir, and Hadrien Bertrand. TorchXRayVision: A library of chest X-ray datasets and models. In *Proceedings of Machine Learning Research (MIDL)*, volume 172, pages 231–249, 2022. URL <https://proceedings.mlr.press/v172/cohen22a.html>.
- [13] Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M Summers. ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2097–2106, 2017. doi: 10.1109/CVPR.2017.369.
- [14] Alistair E W Johnson, Tom J Pollard, Seth J Berkowitz, Nathaniel R Greenbaum, Matthew P Lungren, Chih-ying Deng, Roger G Mark, and Steven Horng. MIMIC-CXR-JPG, a large publicly available database of labeled chest radiographs. arXiv:1901.07042, 2019. URL <https://arxiv.org/abs/1901.07042>.
- [15] Alistair E W Johnson, Lucas Bulgarelli, Lu Shen, et al. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10(1):1, 2023. doi: 10.1038/s41597-022-01899-x.
- [16] Alistair E W Johnson, Lucas Bulgarelli, Tom J Pollard, Steven Horng, Leo A Celi, and Roger G Mark. MIMIC-IV-ED: Emergency department module of MIMIC-IV, 2023. URL <https://physionet.org/content/mimic-iv-ed/>.
- [17] Jeremy Irvin, Pranav Rajpurkar, Michael Ko, et al. CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In *AAAI*, volume 33, pages 590–597, 2019. doi: 10.1609/aaai.v33i01.3301590.

- [18] Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q Weinberger. Densely connected convolutional networks. In *CVPR*, pages 4700–4708, 2017. doi: 10.1109/CVPR.2017.243.
- [19] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *KDD*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.